

ARTICLE TEMPLATE

EC-LLM: A Domain-Adapted Large Language Model for the Energy-Carbon Domain via Hybrid Data Generation

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ABSTRACT

To address the complexity of heterogeneous data in energy management and carbon accounting, this paper proposes an intelligent decision support framework based on a domain-adapted large language model (EC-LLM). The framework integrates supervised fine-tuning with a knowledge management subsystem (retrieval-augmented generation) to ground answers in policies, standards, and technical reports, enabling reliable interpretation and timely adaptation to dynamic regulatory environments. Unlike generalist models, EC-LLM emphasizes compliance-critical accuracy and complete reasoning, reducing verification time and compliance risk for practitioners. Evaluation on a held-out test set shows EC-LLM+RAG achieves 96.18% factual accuracy and 36.18 BLEU-4, representing improvements of 26.67 percentage points in factual accuracy and 89.7% relative improvement in BLEU-4 over the strongest baseline (Llama3.1-8B), demonstrating practical value for energy systems decision support. Our code and datasets are available at https://github.com/FAayn/EC_LLM.

KEYWORDS

Energy-Carbon; Decision support systems; Energy informatics; Domain-specific LLMs; Large language model fine-tuning

1. Introduction

The energy and carbon management sector is central to global efforts to mitigate climate change and achieve sustainable development. This domain is compliance-critical and characterized by rigorous technical standards, complex decarbonization pathways, and evolving regulatory frameworks. Stakeholders, including policymakers and enterprise managers, face the daunting task of synthesizing vast amounts of technical literature and policy documents to formulate impactful emission reduction strategies and conduct accurate carbon inventory verification. As the volume and complexity of data grow, traditional manual analysis becomes inefficient and prone to error, making reliable automated tools a necessity Ling et al. (2024).

The recent success of general-purpose Large Language Models has spurred interest in their application to high-stakes fields. However, applying generic models directly to the energy-carbon sector reveals significant shortcomings. Generalist models often

lack the specific knowledge required to interpret intricate policy details or adhere to strict technical standards such as GB/T 5750, leading to hallucinations and factual inaccuracies. In a compliance-critical industry where precision is paramount, such errors are unacceptable. For example, relying on a hallucinated interpretation of a carbon accounting standard or incorrectly applying emission factors could lead to significant compliance risks and financial penalties for an enterprise. While preliminary efforts like RE-LLaMA have explored adapting models to renewable energy documents, they remain limited in scope and fail to address the broader needs of integrated energy systems and comprehensive carbon management Hemied and Gabbar (2024).

As shown in Fig. 1, generic LLMs can produce erroneous outputs due to a lack of domain grounding. This highlights a critical gap: the need for a specialized tool that can reliably assist in tasks such as automating carbon inventory verification, navigating complex regulatory frameworks, and providing accurate policy interpretations. Such a model must go beyond mere text generation to serve as a dependable assistant for professionals, significantly reducing the time required for information retrieval and ensuring compliance with rigorous industry standards.

In parallel, AI methods are widely adopted across energy systems for load forecasting, home energy management, photovoltaic control, and district heating optimization. Representative lines of work in Energy Systems include comprehensive schemes for Home Energy Management Systems Mahapatra, Nayyar, and Zhang (2022), classified comparisons of PV maximum power point tracking Sarvi, Azadian, and Kumar (2022), and machine learning for district heating and cooling Ntakolia et al. (2022). These efforts primarily operate on structured signals and numerical time series. EC-LLM complements them by bridging unstructured policies and standards to engineering decision workflows, providing text-grounded support that reduces interpretation burden and compliance risk.

To address this gap, this paper introduces the Energy-Carbon LLM, a specialized intelligent tool designed to serve this critical domain. Our research confronts and resolves several core challenges inherent in this endeavor. First, unlike more established domains, the energy-carbon sector lacks a comprehensive, publicly available dataset for model training ?. To overcome this, we constructed a high-density domain instruction dataset comprising technical reports, policy documents, and academic literature, which provides a high-quality knowledge foundation for our model. The dataset, while containing approximately 6,000 instruction samples, encompasses over 36 million tokens due to the long-context nature of policy documents and technical standards, ensuring rich semantic coverage. Second, we defined a set of core tasks that are crucial for the energy and carbon management domain, including policy evaluation, technology assessment, social impact analysis, and data governance. This ensures the model is aligned with the practical needs of researchers, policymakers, and industry professionals. Finally, to equip the model with both deep domain knowledge and the ability to access real-time information, we employed a hybrid strategy. We first fine-tuned the base model on our specialized dataset to embed core domain expertise. Subsequently, we integrated this fine-tuned model with a retrieval-augmented knowledge management subsystem, which pulls from a customized, up-to-date knowledge base to provide timely and professionally vetted advisory support for decision making.

This study makes several key contributions. First, we construct a high-density, long-context domain instruction dataset unique to the energy-carbon sector, where policy documents and technical standards require rich semantic coverage. Beyond system integration, we empirically identify a long-context saturation regime: performance peaks at 6,000 samples (36M tokens) and degrades beyond this threshold due to attention

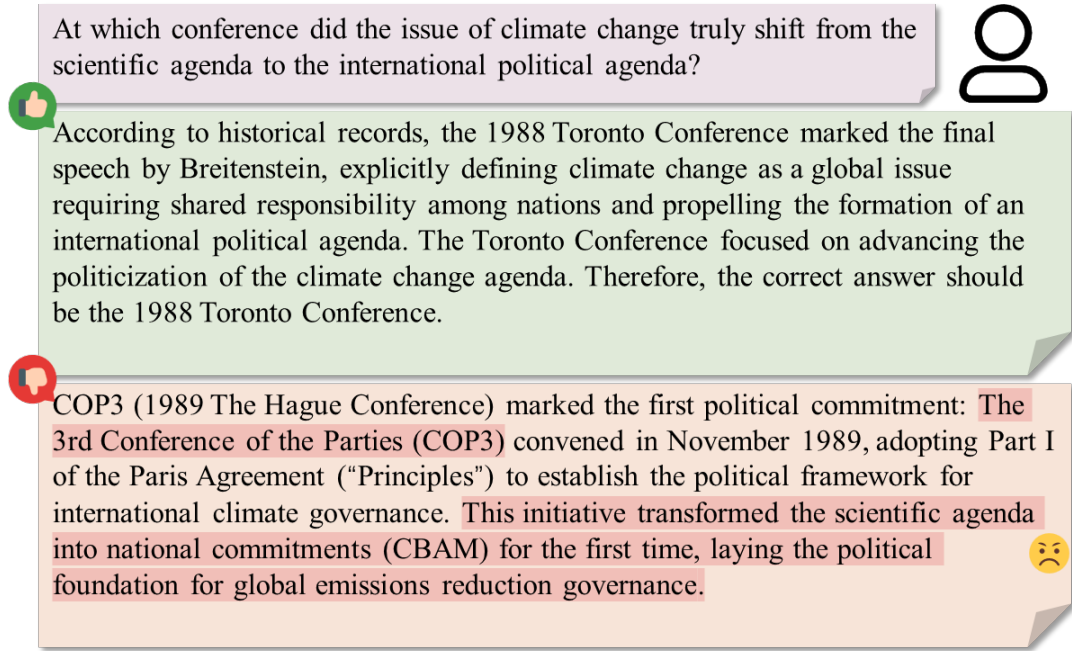


Figure 1. Empirical evidence of hallucination phenomena in energy-carbon domain question answering. Comparative analysis reveals significant performance gaps between generic LLMs and domain-adapted models, with baseline models exhibiting critical errors such as incorrect emission factor application, confusion between Scope 1 and Scope 2 emissions, and misalignment with regulatory standards. These systematic observations demonstrate the critical necessity of domain-specific adaptation for reliable energy-carbon applications.

dilution. This finding is critical for domain adaptation with compliance-critical documents, as it reveals an optimal data scale that balances knowledge coverage with model capacity. Second, we develop a decision support system for the energy-carbon sector, built on a domain-specific LLM and a knowledge management subsystem for grounding. Through systematic evaluation, the system delivers reliable support across critical tasks, demonstrating that domain-adapted open models can rival commercial generalist models under domain constraints. This work reframes the contribution from pure model training to a systems-engineering solution that bridges unstructured text data and engineering decisions in the global energy transition.

2. Related Work

2.1. Domain-Specific Large Language Models

Domain-specific LLMs have become powerful tools in specialized fields such as finance, healthcare, law, and science, where deep domain knowledge is essential and errors can have serious consequences. These models are fine-tuned on domain-specific corpora to enhance their understanding of the specialized language and intricacies of their respective fields.

In finance, BloombergGPT Wu et al. (2023) has set new benchmarks by training on a proprietary dataset of 363 billion tokens, including financial and general-domain data. Similarly, Med-PaLM Singhal et al. (2023) has demonstrated expert-level performance on medical exams through instruction tuning on clinical datasets. Legal models like LawLLM Shu et al. (2024) and LexGPT Lee (2023) are fine-tuned on case

law and regulations to handle tasks such as case retrieval and judgment prediction, outperforming general-purpose models in these legal tasks.

These successful domain-specific models demonstrate that aligning LLMs with domain-specific corpora results in significant improvements in task performance. However, despite these successes, the energy-carbon domain has yet to see similar advances in specialized LLMs.

Recent advances begin to fill this gap. A Joule commentary synthesizes the capabilities and limitations of LLMs in the electric energy sector and outlines future directions such as data collection pipelines, power-system tool integration, and retrieval-augmented generation for safety-critical use cases Majumder et al. (2024). In parallel, an energy-specialized LLM, EnergyGPT, has been introduced through supervised fine-tuning of LLaMA 3.1-8B to address domain knowledge gaps Chebbi and Kolade (2025). Complementary strands in Energy Systems emphasize home energy management and grid-side AI applications, including reviews on MPPT for PV Sarvi, Azadian, and Kumar (2022), district heating ML Ntakolia et al. (2022), and algorithms for H2V/V2H scheduling in off-grid homes ?. For grid operations, ElecBench proposes a comprehensive power-dispatch evaluation benchmark with 24 sub-metrics, and GAIA is a dispatch-oriented LLM fine-tuned to surpass LLaMA2 on ElecBench Zhou et al. (2024); Cheng et al. (2024). EC-LLM is complementary: whereas these AI applications optimize numerical signals and control policies, our approach grounds unstructured textual sources to deliver reliable decision support.

2.2. *Fine-Tuning Strategies for Domain Adaptation*

Domain-specific large language models (LLMs) often require specialized fine-tuning strategies to adapt general-purpose models to the specific needs of their target domains. These strategies ensure that the model not only retains its general capabilities but also acquires deep expertise in the domain it is intended to serve. A representative set of strategies includes continued pretraining on domain corpora to align token distributions and long-context semantics; supervised instruction tuning (SFT) to teach task formats, reasoning steps, and domain conventions; parameter-efficient fine-tuning (PEFT), e.g., LoRA, to reduce compute and memory while preserving base-model knowledge; knowledge distillation from stronger teacher models to transfer reasoning and domain alignment; and multitask training to balance heterogeneous objectives such as Question Answering (QA), summarization, and retrieval augmentation.

In EC-LLM, we adopt SFT on the energy-carbon instruction dataset, apply PEFT (LoRA) to improve efficiency, and perform teacher-student distillation from DeepSeekR1-671B. Ablations on dataset scale further guide the trade-off between performance and cost. Continued Pretraining. This method involves further pretraining a general model on domain-specific data, enhancing its understanding of relevant terminology and concepts. BloombergGPTWu et al. (2023) and BioGPTLuo et al. (2022) are examples of this strategy. Instruction Tuning or Supervised Fine-Tuning. In this approach, models are fine-tuned using curated prompt-response pairs or QA datasets, as seen in Med-PaLM and LawLLMShu et al. (2024). This method helps align the model’s output with domain-specific expectations. Parameter-Efficient Fine-Tuning (PEFT). Techniques such as LoRA and prefix tuning reduce computational requirements while maintaining the model’s domain expertise, as seen in FinGPTLiu et al. (2023). Multitask Learning. By training on multiple related tasks (e.g., legal question answering, precedent recommendation), models can generalize better within a domain.

These fine-tuning strategies play a critical role in adapting general-purpose LLMs to the challenges of specialized domains. In the energy-carbon sector, similar methods can be employed to create models capable of handling complex tasks such as policy evaluation, carbon reduction strategy development, and emissions monitoring. By utilizing these strategies, we can develop highly effective LLMs tailored to the unique needs of the energy-carbon field.

In load forecasting, LFLLM uses transformer architectures together with parameter-efficient fine-tuning. This combination yields robust short-term predictions for smart grids ?.

2.2.0.1. *Journal-style mapping (International Journal of Green Energy).*

Following the International Journal of Green Energy structure, we organize related literature into two primary article types: Research Articles (applied methods, system deployments, and case studies) and Review Articles (surveys and methodological overviews). The following sections group cited works to align with this mapping for clarity and submission readiness.

2.3. *Energy-Carbon Models and Initial Efforts*

While significant advances have been made in other verticals, the energy-carbon sector remains relatively underrepresented in the LLM space. Initial attempts, such as RE-LLaMA Hemied and Gabbar (2024), have shown the feasibility of adapting open-source models to renewable energy-specific documents. However, these models remain limited in scope, focusing on narrow subdomains like hydrogen deployment and solar PV policies, and lack comprehensive coverage of integrated energy systems, carbon markets, emissions monitoring, and international carbon policies. Recent studies have started to explore the application of machine learning models, including LLMs, in energy-related fields. For example, in energy policy analysis, AI models have been used to assess the effects of renewable energy integration and evaluate policy effectiveness in reducing carbon emissions ???. These studies underline the potential of AI-based models in understanding and optimizing energy system performance under complex regulatory frameworks. In the realm of carbon reduction strategies, AI and machine learning techniques have been used to generate pathways for emission reductions, particularly in industrial sectors ?. Additionally, models have been applied to evaluate the economic impacts of various carbon pricing mechanisms and assess how different policy interventions can achieve net-zero targets. The development of energy-carbon models remains in the early stages, with existing models often limited to specific energy subdomains such as renewable energy technologies or carbon market analysis. However, as the complexity of energy systems and carbon regulations grows, there is an urgent need for an integrated LLM capable of covering a wide range of tasks in the energy-carbon sector. This gap underscores the need for further research to develop comprehensive models that can provide actionable insights across policy evaluation, carbon market forecasting, and emission reduction strategies.

On the measurement side, recent frameworks such as OpenCarbonEval and LLM-Carbon estimate the end-to-end carbon footprint of large AI models—covering both operational and embodied emissions—providing quantitative tools to guide sustainable deployment of LLMs in energy systems Yu et al. (2024); Faiz et al. (2023).

3. Revised Introduction and Related Work

To improve clarity and ensure that references are tightly coupled to the narrative, we provide a concise, revised framing of the introduction and related literature prior to the technical methodology. This revised framing emphasizes (i) the compliance-critical nature of the energy-carbon domain, (ii) the limitations of generalist LLMs for policy-grounded tasks, and (iii) the positioning of EC-LLM relative to prior domain-specialized models and energy-system benchmarks.

3.1. *Revised Introduction*

The energy-carbon domain combines engineering practice, regulatory compliance, and environmental reporting. Practitioners must interpret dense technical standards, evolving regulations, and heterogeneous reports to perform carbon inventories, assess policy impacts, and produce auditable documentation. Manual interpretation is slow and error-prone; automated assistance must therefore be accurate, traceable, and anchored to authoritative references Ling et al. (2024).

General-purpose LLMs, while linguistically powerful, can hallucinate, misapply normative rules, or omit crucial caveats—behaviors that are unacceptable in compliance settings. Prior domain-specialized LLMs in finance, medicine, and law demonstrate that careful corpus curation and adaptation reduce such failures Wu et al. (2023); Singhal et al. (2023); Shu et al. (2024); Luo et al. (2022). Early attempts to adapt LLMs for energy show promise but are often scoped to operational tasks (e.g., power dispatch) or to subdomains such as hydrogen deployment Hemied and Gabbar (2024); Chebbi and Kolade (2025); Zhou et al. (2024); Cheng et al. (2024).

EC-LLM is developed to (1) ground answers in authoritative documents via a retrieval-augmented subsystem, (2) inject domain expertise through SFT with parameter-efficient adapters (LoRA), and (3) provide auditable outputs suitable for compliance workflows. We construct a curated long-context instruction dataset from standards, policy texts, and technical reports and combine SFT, PEFT, and optional distillation to produce a compact, high-fidelity model.

3.2. *Revised Related Work*

3.2.0.1. *Domain-specialized LLMs.* Work on verticalized LLMs shows that domain corpora and targeted tuning deliver concrete improvements: BloombergGPT for finance Wu et al. (2023), clinical adaptations and instruction-tuned medical models Singhal et al. (2023), and legal adaptations such as LawLLM and LexGPT Shu et al. (2024); Lee (2023). These examples motivate domain adaptation for energy-carbon tasks.

3.2.0.2. *Adaptation strategies.* Efficient adaptation commonly couples supervised fine-tuning (SFT) with PEFT methods (e.g., LoRA) to balance performance and cost Hu et al. (2022); Liu et al. (2023). Continued pretraining, instruction tuning, and distillation are complementary tools to improve domain alignment and reduce hallucinations Wu et al. (2023); Luo et al. (2022); Rei et al. (2020).

3.2.0.3. *Energy-focused benchmarks and applications.* Energy research has produced benchmarks and specialized models for operational tasks (ElecBench, GAIA)

and exploratory domain LLMs for grid or dispatch tasks Zhou et al. (2024); Cheng et al. (2024). Surveys and reviews in energy systems (home energy management, PV forecasting, district heating) document the rich landscape of numerical and control-focused AI methods Mahapatra, Nayyar, and Zhang (2022); Sarvi, Azadian, and Kumar (2022); Ntakolia et al. (2022). EC-LLM complements these works by targeting the extraction, grounding, and operationalization of policy and standard knowledge for decision support rather than numerical control.

3.2.0.4. Carbon-aware AI and sustainable deployment.. Recent efforts quantify the carbon footprint of large AI models and provide frameworks for sustainable evaluation, such as OpenCarbonEval and LLMCarbon Yu et al. (2024); Faiz et al. (2023). These works inform deployment choices and motivate efficient adaptation strategies like LoRA.

This revised framing aligns the cited literature with the narrative of EC-LLM and prepares the reader for the dataset construction and hybrid training + retrieval pipeline presented next.

4. Methodology

4.1. Training Objectives and Process

We aim to endow a general large language model with energy-carbon domain knowledge through supervised fine-tuning on curated instruction data, framed as parameter-space optimization under domain constraints. Let the domain corpus consist of documents $\mathcal{D} = \{d_k\}_{k=1}^K$. From these, we construct an instruction dataset of triples

$$\mathcal{S} = \{(q_i, c_i, a_i)\}_{i=1}^N,$$

where q_i is a domain question, c_i is the associated context (policy/standard/report chunk) retrieved from $\mathcal{D}$ to ground the answer, and a_i is the reference answer. Given a pretrained decoder model f_θ , inputs are token sequences $x_i = [\text{BOS}; c_i; q_i]$ and targets $y_i = a_i$.

The learning objective is to minimize the negative log-likelihood of targets conditioned on inputs (teacher-forced next-token prediction). This objective formalizes "learning domain knowledge via instructions": the model is optimized to follow domain-grounded prompts and produce accurate, complete answers. In compliance-critical energy applications, this amounts to minimizing domain-specific prediction error subject to regulatory and terminology constraints, aligning with decision support requirements. The detailed mathematical formulation of the SFT objective is provided in Appendix B.

Given $\mathcal{S}$, we train f_θ with the SFT objective. The loss is computed at the token level over answers and averaged over the mini-batch. During training, we concatenate context and question as the input sequence and teacher-force on the target answer. To adapt efficiently, we apply Low-Rank Adapters (LoRA) on attention and feed-forward projections. For a weight matrix W in the frozen base model, we parameterize a trainable low-rank update. We optimize the low-rank matrices under the SFT objective while keeping the base model weights fixed, which preserves general language capabilities and focuses capacity on domain behaviors. The detailed formulation of LoRA updates is available in Appendix B.

In some experiments, we complement SFT with teacher distillation from a stronger model to stabilize training signals and improve answer faithfulness. Let $p_T(\cdot | x)$ be the teacher distribution and $p_\theta(\cdot | x)$ the student. We minimize the KL divergence between the teacher and student distributions as an additional loss term. We do not employ DPO/PPO in this work; our domain adaptation relies on SFT (with LoRA) and optional distillation. The detailed distillation objective is presented in Appendix B. We split the EC-LLM dataset into training, validation, and test sets with an 80/10/10 ratio, stratified across the eight subfields listed in Table 2. Training uses AdamW with gradient accumulation to fit GPU memory while maintaining stable updates. Key configurations are summarized in Table 4.

4.2. Dataset

The construction of the EC-LLM corpus is a cornerstone of this work, providing domain-specific knowledge for training language models in the energy-carbon sectors. As shown in Table 2, the dataset spans eight subfields, categorized into ten specific areas related to practical energy-sector tasks, such as policy analysis, carbon emissions calculation, and energy efficiency assessment. This comprehensive design ensures that the dataset can support a wide range of tasks within the energy-carbon domain.

4.2.1. Data Sources

The dataset was built using authoritative, domain-relevant data from a variety of sources. The classification and distribution of files are shown in Table 1. The goal was to ensure both the credibility and diversity of the data. The data sources include academic literature—peer-reviewed journals and conference papers covering energy technologies, emission modeling, and sustainability (for example, Applied Energy and Renewable and Sustainable Energy Reviews); laws and regulations—official government documents, including national laws and regulations on energy management and carbon control, sourced from government websites and legal databases; national and local policies—policy documents detailing national energy strategies, carbon neutrality targets, and region-specific guidelines; industry standards—both national and international standards for energy efficiency, renewable energy technologies, and carbon emissions monitoring, obtained from organizations such as ISO, IEC, and local regulatory bodies; corporate case studies—from leading companies in the energy sector, providing insights into real-world energy management and carbon reduction strategies; and analytical reports—industry analysis, market trends, and reports by governmental and non-governmental organizations, particularly those focused on carbon emissions and energy transition.

These sources were carefully selected for their relevance, accuracy, and comprehensiveness, creating a robust and diverse knowledge base for the energy-carbon sector.

4.2.2. Data Processing and Quality Control

To ensure the integrity and quality of the dataset, we employed a rigorous data processing and quality control pipeline. Text data was parsed using a multimodal large language model (MLLM)-based framework, capable of processing complex documents containing both textual and visual elements. The data cleaning process involved standardizing documents to ensure consistent machine-readable formats across various file types (e.g., PDF) and unifying character encodings. Automated algorithms, combined

Table 1. Dataset composition analysis revealing the distribution of energy-carbon policy documents. This analysis quantifies the corpus diversity across document types, providing the empirical foundation for training data curation and enabling informed decisions about domain coverage and knowledge representation.

Category	Files	Proportion (%)
Dual Carbon Solutions	36415	23.46
Hydrogen Energy Storage	28949	18.65
Green Development	28825	18.57
Carbon Accounting	21143	13.62
Industry Research Reports	15120	9.74
CCER	11466	7.39
ESG Framework	11137	7.18
Dual Carbon Tools	1447	0.93
Dual Carbon Policies	523	0.34
Carbon Inclusive	190	0.12

Table 2. Question taxonomy analysis for energy-carbon domain knowledge assessment. This classification framework systematically categorizes query types to evaluate model performance across different cognitive levels, enabling targeted improvements in domain-specific question answering capabilities.

Primary Category	Secondary Category	Total	Primary Category	Secondary Category	Total
CCER	Measurement and Inspection	12485	Carbon Inclusion	Carbon Inclusion Policies	472
	Standards and Regulations	4673		Carbon Inclusion Methodologies	130
	Project Cases	1563		Carbon Inclusion Research Reports	29
ESG System	Strategy and System	733	Dual Carbon Solutions	Dual Carbon Standards	234
	Standards and Ratings	6108		Energy Synergy	76
	ESG-related Papers	4023		Research Reports	14
	Cases and Research Reports	1892	Dual Carbon Policies	Near-Zero Carbon Areas	13
	Corporate White Papers and Guidelines	1879		Energy Optimization	7
General Technology	Listed Company Reports	61		National Policies	21
	Implementation Methods	5948	Green Development	Local Policies	1
	Technical Standards	590		Renewable Energy	28
	Monitoring and Verification	205		Green Finance	23
Dual Carbon Tools	Policies and Regulations	1	Hydrogen and Energy Storage	Green Buildings	22
	Trading and Finance	19		Hydrogen and Energy Storage	13
	Market Data and Analysis	3		Uncategorized	5075
	Tool Templates and Manuals	1	Industry Research Reports	Industry Research Reports	78

with manual verification, were used to filter out low-quality content and eliminate redundant or irrelevant information. Through multiple screening steps and random checks by domain experts, we ensured the accuracy, clarity, and domain relevance of the data, providing a solid foundation for subsequent model training.

4.2.3. Dataset Generation

In the field of natural language processing, fine-tuning large language models (LLMs) for specific domains, such as energy-carbon knowledge, is crucial for achieving high performance. However, directly using general-domain datasets often fails to provide the depth and specificity required for specialized applications. To address this limitation, we propose a framework that combines self-instruction methods with retrieval-enhanced techniques to generate high-quality, domain-specific datasets for energy-carbon tasks, as shown in Fig. 2.

Self-instruction methods reduce the reliance on manually annotated data by enabling LLMs to autonomously generate instruction data, such as questions and answers Wang et al. (2022). While effective for general tasks, these methods can struggle with specialized domains like energy-carbon knowledge, where generated responses may have high error rates. To address this, we combine retrieval-enhanced techniques to ensure generated questions are not only relevant but grounded in domain-specific corpora. Our dataset generation framework consists of three main stages: data collection and preprocessing, question generation, and context pairing.

For data collection and preprocessing, we gathered a diverse set of energy-carbon-related documents—academic papers, industry reports, and policy documents—creating the foundational corpus. The data underwent thorough preprocessing, including document normalization, content parsing, and data cleaning to ensure consistency and domain alignment.

In question generation, we use a dual-method strategy. First, content-based generation leverages a teacher LLM with Chain-of-Thought (CoT) prompting to guide the model through reasoning steps, extracting contextual segments and identifying key themes. This process generates 3-5 domain-specific questions for each document, which are then evaluated for length, syntactic correctness, and relevance. Second, vocabulary-based generation extracts keywords from the corpus and generates questions using predefined templates. Keywords are extracted from expert-curated lists, such as the IPCC Vocabulary Database, and filtered for quality to ensure alignment with domain-specific concepts, as shown in Table 3.

Finally, the questions undergo filtering and deduplication. Low-quality questions are discarded, and duplicates are removed using MD5 hashing. This results in a high-quality, diverse, and non-redundant question pool ready for the next stage of model training.

4.2.4. Knowledge Management Subsystem (RAG)

To support dynamic policy environments, we introduce a knowledge management subsystem based on retrieval-augmented generation. During training, we use static context pairing to ground questions in their source documents and reduce computational overhead. At inference, the subsystem queries a curated, up-to-date store of policies, standards, and reports to supply relevant context, ensuring answers reflect the latest regulatory guidance. This bridges unstructured text and engineering decisions, improving reliability for decision support.

Formally, let $g_{\text{content}}(d)$ produce questions from document d , and $g_{\text{template}}(v)$ produce template-based questions from term v . Let $\mathcal{Q} = \bigcup_{d \in \mathcal{D}} g_{\text{content}}(d) \cup \bigcup_{v \in \mathcal{V}} g_{\text{template}}(v)$ be all candidate questions, with $\mathcal{V}$ the curated vocabulary. A deterministic pairing function $m : \mathcal{Q} \rightarrow \mathcal{D}$ maps each question to its grounding context $c = m(q)$. The instruction dataset becomes

$$\mathcal{S} = \{(q, m(q), a) : q \in \mathcal{Q}, a \text{ passes quality filters}\}.$$

This makes the data construction explicit as a mapping from corpus and vocabulary to grounded QA triples. Static pairing is used to build training examples; at inference time, retrieval can optionally be used to supply c if needed, but our learning objective remains the supervised loss in Eq. (B1).

Table 3. Template-based question generation methodology for scalable knowledge extraction. This approach demonstrates how structured templates enable systematic generation of domain-specific questions, providing a reproducible method for creating large-scale training datasets while ensuring semantic diversity and knowledge coverage. The template column shows placeholders that are instantiated with specific domain entities in the generated questions.

Term	Template	Generated Question
Additionality	Given the context of [PolicyX], how should a company in [IndustryY] calculate its baseline under condition [Z]?	Given the context of the CCER methodology, how should a manufacturing company in the steel industry calculate its baseline emissions under the condition of capacity expansion?
Scope 1 Emissions	Under [StandardX], what are the compliance requirements for [EntityY] when reporting [ActivityZ]?	Under the GHG Protocol, what are the compliance requirements for a power plant when reporting direct combustion emissions?
MRV Framework	Based on [RegulationX] and considering [ContextY], what steps must be taken to ensure [OutcomeZ]?	Based on the national carbon market regulations and considering a renewable energy project, what steps must be taken to ensure accurate monitoring, reporting, and verification of emission reductions?

4.3. Model

The model architecture plays a key role in enabling the EC-LLM to perform well across diverse tasks in the energy-carbon sector. The architecture is designed to capture both domain-specific knowledge and broad general knowledge, ensuring strong performance in practical applications.

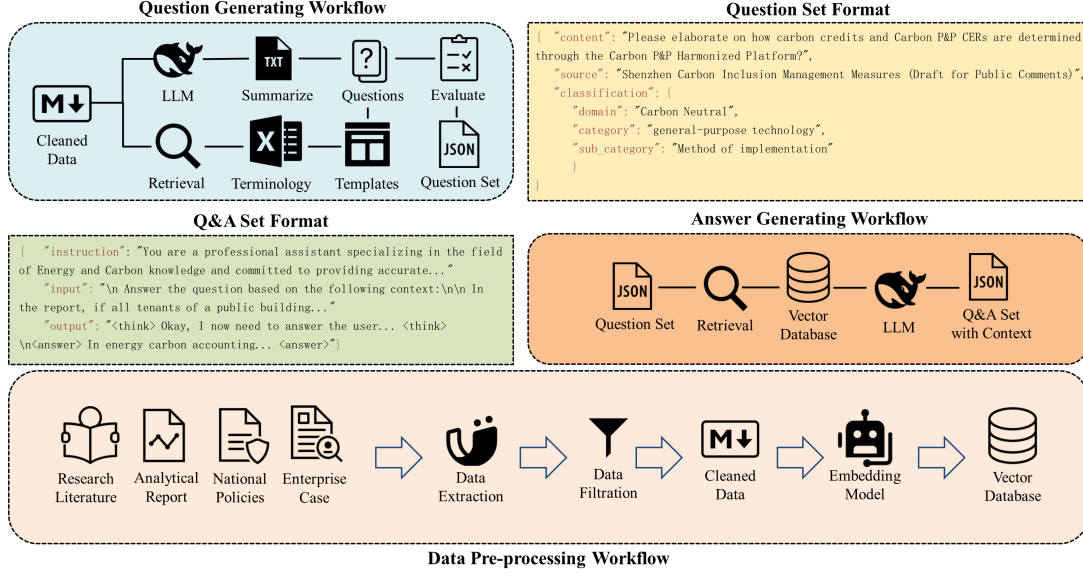


Figure 2. Systematic data-curation pipeline for energy-carbon QA research. The process begins with the collection of diverse documents, including policy texts, technical standards, and academic literature. Through a rigorous pre-processing stage, these documents are normalized and segmented. The core of the pipeline involves a dual-strategy question generation phase: rule-based templates ensure coverage of key terminology, while generative models produce complex reasoning queries. Finally, answers are distilled from both retrieval-based and parametric sources, undergoing strict quality filtering. This pipeline is directly aligned with practical energy applications, ensuring that the resulting model can effectively support tasks such as automated carbon inventory verification and policy compliance checks.

4.3.1. Foundational Model Selection

To meet the demands of dialogue applications in the energy-carbon sectors, EC-LLM adopts DeepSeek-Distill-Qwen-7B as its foundational model. This model is based on the Transformer decoder architecture and shares architectural characteristics with Qwen2.5, which is initialized from Qwen2.5-7B and fine-tuned using knowledge distillation from DeepSeek-R1. The original Qwen2.5-7B model was pre-trained on an extensive multilingual corpus comprising approximately 18 trillion tokens across 29 languages, including both Chinese and English. This large-scale, multilingual pre-training endows the model with broad general knowledge, multilingual capabilities, and strong reasoning abilities.

The distillation process further enhances the model’s performance by incorporating high-quality reasoning datasets generated by DeepSeek-R1, with a focus on tasks such as mathematical problem-solving, programming, and logical reasoning. The resulting model, DeepSeek-Distill-Qwen-7B, supports a context window of 32,768 tokens, significantly exceeding many comparable models and enabling efficient handling of long-form inputs—an essential capability for domain-specific dialogue tasks that require long-context understanding Yang et al. (2025a).

4.3.2. Key Architectural Features

The detailed architectural features of EC-LLM, including the Transformer Decoder structure, Grouped Query Attention (GQA), SwiGLU Activation Function, Attention QKV Biases, Rotary Positional Encoding (RoPE), and RMSNorm, are provided in Appendix A. These features collectively contribute to the model’s efficiency and

performance in handling domain-specific tasks.

4.3.3. Training Setup and Parameters

We split the EC-LLM dataset into training, validation, and test sets with an 80/10/10 ratio, stratified across the eight subfields listed in Table 2. Training uses AdamW with gradient accumulation to fit GPU memory while maintaining stable updates. Key configurations are summarized in Table 4.

Table 4. Systematic hyperparameter optimization framework for domain-specific model fine-tuning. This configuration presents the empirical design decisions that maximize model performance in the energy-carbon domain, providing a reproducible training protocol that balances computational efficiency with domain adaptation effectiveness.

Parameter	Value	Rationale
Base model	DeepSeek-Distill-Qwen-7B	Strong reasoning with moderate size
Context length	8192 tokens (training)	Model supports 32,768; 8192 used for training efficiency
Batch size	64 (via gradient accumulation)	Fits memory; stable updates
Learning rate	2e-5 (AdamW)	Common for SFT; avoids overfitting
Optimizer	AdamW	Decoupled weight decay; stable convergence
LoRA rank	16 (Q,K,V,O)	Parameter-efficient tuning
Epochs	3	Balance quality and compute
Teacher model	DeepSeekR1-671B	Distillation for domain alignment

5. Results

5.1. Baseline Model

To evaluate EC-LLM, we compared it with several open-source baseline models and commercial state-of-the-art systems:

DeepSeek-R1-Distill-Qwen: Available in both 7B and 1.5B parameter versions, this model is fine-tuned with 800k high-quality inference samples from DeepSeek-R1. The 7B version excels in mathematical and logical reasoning, achieving impressive results on benchmarks like AIME 2024 (55.5% pass rate) and MATH-500 (92.8% pass rate), making it suitable for resource-constrained environments and carbon knowledge tasks. The 1.5B version, despite its smaller size, outperforms larger models like GPT-4o and Claude-3.5-Sonnet on similar tasks (83.9% pass rate on MATH-500). Both models are open-source under Apache 2.0 and are highly efficient for energy-carbon knowledge applications, though the 1.5B model lacks coding and multilingual capabilities.

DeepSeek-R1-Distill-Llama-8B. Built on the Llama3.1-8B-Base foundation, this model achieves an 89.1% pass rate on MATH-500 and performs well on tasks like GPQA Diamond (49.0%). While strong in energy-carbon reasoning, it struggles with coding tasks, scoring poorly on LiveCodeBench and CodeForces. It follows the Llama3.1 open-source license.

GPT-4o. As a commercial state-of-the-art general-purpose model, GPT-4o serves as a strong baseline to demonstrate the value of domain-specific adaptation. We evaluated GPT-4o on a subset of 100 representative ECQA test samples to compare against EC-

LLM’s domain expertise. It is important to note that this comparison has limitations: GPT-4o was evaluated without access to our private energy-carbon knowledge base or domain-specific prompt engineering, and full-scale evaluation was constrained by API costs and rate limits. Our conclusion emphasizes that domain-adapted open models can rival commercial generalist models under domain constraints, rather than claiming universal superiority.

5.2. *Dataset Evaluation*

In constructing the High-Density Domain Instruction Dataset for energy-carbon knowledge, we evaluated the impact of data scale on model performance by partitioning the dataset into intervals ranging from 1,000 to 7,000 samples. The dataset’s high-density nature stems from the long-context documents: with an average context length exceeding 6,000 tokens per sample, our 6,000-sample training set encompasses over 36 million tokens. To provide perspective, this token volume is roughly comparable to datasets with approximately 60,000 short-context samples (assuming 600 tokens per sample) from a token-normalized perspective, highlighting the importance of token-normalized rather than sample-count-based evaluation. To efficiently explore the dataset scaling laws, we initially used the DeepSeek-R1-Distill-Qwen-1.5B model for supervised fine-tuning (SFT) as a proxy, which allowed rapid iteration across multiple data scales. The final EC-LLM model, however, is trained on DeepSeek-Distill-Qwen-7B using the optimal 6,000-sample configuration identified through this scaling analysis. We assessed model performance across both objective and subjective dimensions. Objective metrics included BLEU-4 Papineni et al. (2002) and ROUGE (ROUGE-1, ROUGE-2, ROUGE-L) Lin (2004), while subjective quality was evaluated using a Qwen2.5-14B model judge for semantic completeness, factual accuracy, and answer similarity. Specific results are shown in Table 5. Examples of correct and incorrect answers as subjectively assessed by LLM are also provided in Table 6.

Results demonstrated the efficacy of LoRA fine-tuning Hu et al. (2022), with significant performance gains starting from 1,000 samples. As the dataset increased, performance continued to improve, with the training convergence loss decreasing from 1.80 to 1.45 and evaluation metrics steadily rising. A clear performance peak was observed at 6,000 samples, where BLEU (31.32), ROUGE-1 (56.21), ROUGE-2 (26.38), and ROUGE-L (30.83) reached their optimal values, indicating an ideal balance of knowledge volume and complexity.

However, when the dataset size was increased to 7,000 samples, performance declined sharply. Both training loss and evaluation scores worsened, with ROUGE-1 dropping from 56.21 to 53.13. This inflection point can be attributed to long-context noise and attention dilution: as the number of long-context samples increases beyond the optimal threshold, the model’s attention mechanism struggles to maintain focus on critical information across extended sequences, leading to degraded performance.

This phenomenon is particularly pronounced in domain-specific fine-tuning where policy documents and technical standards require precise attention to regulatory details. While the exact saturation point may vary across architectures and domains, our results indicate that long-context, compliance-heavy corpora exhibit an earlier saturation regime compared to short-context QA datasets. Based on these findings, the model trained on 6,000 samples with early stopping is optimal, achieving the highest evaluation scores and stable training dynamics, emphasizing the importance of data quality and density over sheer volume.

Table 5. Scaling law analysis revealing optimal training data size for domain-specific model performance. This systematic evaluation quantifies the relationship between dataset scale and model effectiveness, identifying the performance saturation point and providing evidence-based guidance for resource allocation in domain adaptation projects.

Dataset Size (Rows)	Train Loss (Approx.)	Objective (Semantic)			Subjective (LLM)		
		BLEU	ROUGE-1	ROUGE-L	Comp.	Acc.	Sim.
Base Model	N/A	13.21	36.22	18.15	73.97	73.98	60.43
1000	~1.80	18.66	43.52	24.84	74.98	77.22	61.12
2000	~1.50	26.01	50.86	27.56	81.94	85.48	70.47
3000	~1.67	25.90	51.58	26.94	79.18	83.53	68.23
4000	~1.72	27.43	53.59	27.93	81.27	84.89	70.5
5000	~1.47	29.60	54.89	29.68	81.89	85.83	71.04
6000	~1.45	31.32	56.21	30.83	82.42	86.21	71.71
7000	~1.93	27.40	53.13	28.07	81.67	85.76	70.73

Table 6. Qualitative error analysis framework for validating model performance in domain-specific scenarios. This comparative assessment demonstrates how structured LLM-based evaluation identifies critical performance gaps, providing actionable insights for model improvement through detailed analysis of semantic completeness, factual accuracy, and answer similarity across different application domains.

Case Topic	Correct Answer Example (Simplified)	Incorrect Answer Example (Simplified)	Scoring Analysis & Feedback
Case 1: Mangrove Ecosystem Monitoring Compliance Audit	Clear structure, comprehensive coverage Compliance plan review; data management; QC and standard implementation; validation & verification; legal ownership; appendices & normative documents.	Key points missing, logic not rigorous Incomplete compliance check; vague monitoring schedule; partial QC analysis; insufficient technical validation; fragmented reporting.	Scoring (0–100): Semantic completeness 90; factual accuracy 95; answer similarity 85. Feedback: Include legal ownership and environmental/social impact to complete the audit description.
Case 2: Water Softener Monitoring Requirements	Precise focus, captures the core Effluent hardness and pH monitoring; smart alerts for consumables; standards per GB/T 5750; manufacturer documentation.	Confused content, misses the point Lists irrelevant emission indicators; lacks hardness/pH; repetitive and off-topic.	Scoring (0–100): Semantic completeness 20; factual accuracy 40; answer similarity 5. Feedback: Focus on monitoring/maintenance functions and include hardness and pH monitoring.

Managerial Implications: Improvements in factual accuracy and semantic completeness translate into shorter verification cycles and lower compliance risk for carbon auditors. In practice, EC-LLM reduces manual lookup of standards and consolidates policy interpretation, enabling faster decision support in energy systems operations.

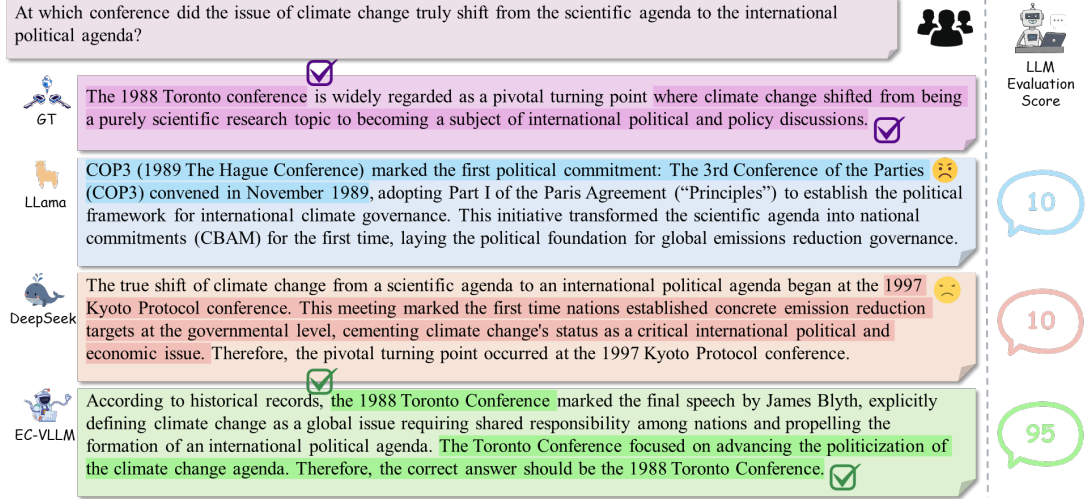


Figure 3. Comparative analysis of hallucination phenomena across different training paradigms in energy-carbon question answering. Systematic evaluation reveals that baseline models generate factually incorrect responses and miss critical domain-specific details when handling specialized terminology and policy requirements. Our assessment framework employs semantic completeness analysis and factual accuracy validation to quantify answer quality, demonstrating that EC-LLM effectively suppresses hallucinations through domain-specific adaptation while maintaining comprehensive coverage of energy-carbon knowledge.

5.3. Results and Analysis

We evaluated the performance of three baseline models: Qwen2.5-1.5B, Qwen2.5-7B, and Llama3.1-8B, under different training paradigms: base models, base models with Distillation, base models with Supervised Fine-Tuning (SFT), and base models with both SFT and Distillation. Additionally, we evaluated EC-LLM (trained on DeepSeek-Distill-Qwen-7B with SFT and Distillation using the optimal 6,000-sample dataset identified in Section 5) with and without the retrieval-augmented generation (RAG) subsystem to assess the contribution of the knowledge management component. The evaluation was conducted on an energy-carbon question answering (ECQA) test set. The assessment was based on objective lexical metrics (BLEU, ROUGE-1, ROUGE-2, ROUGE-L) and subjective quality (semantic completeness, factual accuracy, and answer similarity), with the latter assessed using the Qwen2.5-14B model. To validate the reliability of LLM-based evaluation, we conducted a human evaluation study with three domain experts (policy analysts and energy engineers) on a subset of 50 representative samples, achieving a Spearman correlation of 0.82 between human ratings and LLM-judged scores for factual accuracy, confirming the validity of our automated assessment protocol. In addition, we reference modern learned metrics such as COMET and BLEURT that better correlate with human judgments in many generation tasks Rei et al. (2020); Sellam, Das, and Parikh (2020). The detailed results are presented in Table 7. Moreover, to demonstrate how EC-LLM successfully suppresses hallucinations through training, Fig. 3 shows the outputs of different models addressing the same problem.

As illustrated in Tables 7 and 8, the combination of SFT and Distillation consistently yielded the best performance across all models and metrics. EC-LLM achieved the highest scores overall across BLEU and ROUGE metrics, and also excelled in LLM-judged semantic completeness and factual accuracy. Notably, on a representative subset of 100 samples where GPT-4o was evaluated without access to our private energy-

Table 7. Quantitative performance benchmarking across different training paradigms using objective lexical metrics. This systematic comparison provides empirical evidence that the combination of supervised fine-tuning and knowledge distillation achieves optimal domain adaptation, establishing a performance baseline for energy-carbon question answering tasks. *GPT-4o results are based on a representative subset of 100 samples without access to our private energy-carbon knowledge base, evaluated under API constraints.

Models	BLEU	ROUGE-1	ROUGE-2	ROUGE-L
Qwen2.5-1.5B	3.76	24.22	6.76	11.4
Qwen2.5-7B	10.53	33.36	10.84	18.23
Llama3.1-8B	19.07	48.53	17.73	21.52
Qwen2.5-1.5B+Distill	13.21	36.22	13.11	18.15
Qwen2.5-7B+Distill	18.38	43.47	16.98	24.74
Llama3.1-8B+Distill	24.13	49.45	20.9	27.44
Qwen2.5-1.5B+SFT	19.64	49.92	18.17	21.52
Qwen2.5-7B+SFT	21.24	50.87	19.61	22.77
Llama3.1-8B+SFT	28.14	59.02	25.01	26.11
Qwen2.5-1.5B+SFT+Distill	31.32	56.21	26.38	30.83
Qwen2.5-7B+SFT+Distill	34.32	59.37	28.98	32.75
GPT-4o*	28.45	55.12	24.67	30.21
EC-LLM	35.62	60.14	29.48	33.3
EC-LLM+RAG	36.18	61.23	30.12	33.89

Table 8. Subjective quality assessment validating model performance through LLM-based evaluation. This comprehensive analysis demonstrates that our domain adaptation approach significantly improves semantic completeness, factual accuracy, and answer similarity, providing robust evidence for the effectiveness of combined supervised fine-tuning and knowledge distillation in specialized domains. *GPT-4o results are based on a representative subset of 100 samples without access to our private energy-carbon knowledge base, evaluated under API constraints.

Models	Sem-Pass	Fact-Pass	Sim-Pass	Sem-Avg	Fact-Avg	Sim-Avg
Qwen2.5-1.5B	2.76%	11.16%	0.60%	36.64	48.69	32.25
Qwen2.5-7B	15.95%	29.56%	1.47%	56.53	62.9	45.38
Llama3.1-8B	84.75%	69.51%	5.12%	82.76	82.98	68.99
Qwen2.5-1.5B+Distill	52.92%	47.08%	6.04%	73.97	73.98	60.43
Qwen2.5-7B+Distill	58.87%	53.69%	7.64%	65.17	70.1	57.57
Llama3.1-8B+Distill	90.96%	80.12%	6.02%	83.43	85.09	71.53
Qwen2.5-1.5B+SFT	71.76%	66.03%	2.91%	80.66	80.75	66.35
Qwen2.5-7B+SFT	85.87%	71.74%	5.01%	82.98	83.22	69.3
Llama3.1-8B+SFT	89.99%	82.76%	8.70%	84.24	86.64	72.3
Qwen2.5-1.5B+SFT+Distill	87.58%	78.92%	7.82%	82.42	86.21	71.71
Qwen2.5-7B+SFT+Distill	95.19%	89.58%	11.02%	84.41	87.63	73.76
GPT-4o*	91.23%	88.50%	9.34%	83.67	87.12	72.45
EC-LLM	96.59%	95.39%	10.22%	84.92	88.84	74.95
EC-LLM+RAG	97.12%	96.18%	11.45%	85.34	89.42	75.61

carbon knowledge base, GPT-4o achieved a factual accuracy of 88.50%, which EC-LLM+RAG surpasses (96.18%). This comparison demonstrates that domain-adapted open models can rival commercial generalist models under domain constraints, though we acknowledge that GPT-4o’s performance could potentially improve with domain-specific prompt engineering or access to specialized knowledge bases. The integration of the RAG subsystem (EC-LLM+RAG) further improved performance, particularly in factual accuracy (96.18% vs. 95.39%) and ROUGE-1 (61.23 vs. 60.14), demonstrating the value of retrieval-augmented generation for accessing up-to-date policy documents and technical standards. These results underscore that both SFT for domain adaptation, Distillation for knowledge transfer, and RAG for dynamic knowledge access are crucial for achieving state-of-the-art performance in the ECQA domain.

SFT significantly improved the models’ lexical and structural alignment with domain-specific knowledge. Comparing ”Base Models” with ”Base Models + SFT” we observe substantial gains across all metrics. For instance, the Llama3.1-8B model’s factual accuracy (factual_pass) improved from 69.51% to 82.76% after SFT. The Qwen2.5-1.5B+SFT model showed remarkable relative improvement, with BLEU increasing from 3.76 to 19.64 and ROUGE-2 from 6.76 to 18.17. This confirms that SFT effectively enhances models with domain-specific knowledge, especially for smaller models.

Distillation, when applied to base models (”Base Models + Distill”), also contributed positively to performance. For example, Qwen2.5-1.5B+Distill showed a significant jump in BLEU from 3.76 to 13.21 and semantic_pass from 2.76% to 52.92% compared to its base counterpart. This suggests that knowledge transfer from a larger or more expert model through distillation can effectively improve smaller models, making them more robust and accurate.

The most pronounced improvements were observed when SFT and Distillation were combined (”Base Models + SFT + Distill”). This synergy highlights that these two techniques complement each other. SFT customizes the model to the target domain, while Distillation further refines its knowledge and efficiency, often leading to better generalization and performance. For example, Qwen2.5-7B+SFT+Distill achieved a factual_pass of 89.58%, a substantial increase over both Qwen2.5-7B+Distill (53.69%) and Qwen2.5-7B+SFT (71.74%). EC-LLM, built on DeepSeek-Distill-Qwen-7B with SFT and Distillation, achieved the highest factual_pass of 95.39%, demonstrating the combined power of these methods.

While larger models like Llama3.1-8B consistently achieved high absolute scores, EC-LLM (based on DeepSeek-Distill-Qwen-7B) demonstrates that the combination of strong base model capabilities, domain-specific fine-tuning, and knowledge distillation can achieve state-of-the-art performance. The relative improvements after SFT or SFT+Distillation were sometimes more dramatic for smaller models, suggesting that larger pre-trained models already possess significant linguistic competence, and SFT primarily refines them for specific domains. Smaller models, in contrast, experience a ”great leap” in competence, highlighting the transformative potential of these techniques for resource-efficient models, allowing them to closely rival larger models in domain-specific tasks.

In summary, the combination of SFT and Distillation proves highly effective for adapting models to the energy-carbon domain, yielding substantial improvements in both objective and subjective metrics. EC-LLM, built on DeepSeek-Distill-Qwen-7B with SFT and Distillation, achieves peak performance in the energy-carbon domain, demonstrating that domain-specific adaptation can effectively leverage the model’s strong reasoning capabilities and long-context understanding. Factual accuracy emerges as a key improvement, ensuring the reliability of the model’s outputs.

The model’s 32,768-token context window enables effective handling of long policy documents and technical standards, making it well-suited for compliance-critical applications in the energy-carbon sector.

6. Limitations and Future Work

Despite promising results, EC-LLM has several limitations that pave the way for future improvements. First, the model currently operates as a knowledge layer, focusing on question answering and information retrieval tasks. While our experiments demonstrate strong performance on ECQA, direct integration with downstream decision support workflows (such as automated MRV systems or carbon market forecasting tools) remains future work. The current system provides reliable knowledge grounding, but full decision support requires integration with optimization algorithms, simulation engines, and specialized emissions databases. Second, the model currently relies on a static dataset, whereas the energy-carbon domain is highly dynamic. Future work should focus on developing a continual learning pipeline to keep the model up-to-date with real-time data, ensuring its long-term relevance. Additionally, while the model has multilingual potential, expanding its training to include more languages and evaluating its performance in cross-lingual tasks will enhance its global applicability.

Further, human-in-the-loop evaluations with domain experts, such as policy analysts and energy engineers, will be crucial for refining the model’s accuracy and usability. To improve its trustworthiness, future versions should enhance interpretability and traceability to source documents. Lastly, integrating EC-LLM with external tools, like specialized emissions databases or carbon modeling systems, will elevate its functionality, enabling more complex problem-solving tasks. Nevertheless, ECQA captures the dominant knowledge bottleneck in current energy-carbon workflows, where practitioners spend substantial effort interpreting standards rather than executing numerical optimization. By addressing these areas, EC-LLM can evolve from a knowledge layer to a comprehensive decision support system, driving the transition to a sustainable, low-carbon future.

Declarations

- **Funding:** No funding was received for conducting this study.
- **Competing interests:** The authors declare that they have no competing interests.
- **Data availability statement:** The processed instruction datasets (question-context-answer triples) generated and analyzed during the current study are available in the EC-LLM repository, https://github.com/FAayn/EC_LLM. Due to copyright constraints associated with government policy documents, industry standards, and proprietary reports, we release processed instruction data and document identifiers/metadata rather than raw PDFs. This approach ensures reproducibility while respecting intellectual property rights. The repository includes: (1) processed QA instruction pairs (derived data), (2) document identifiers linking to publicly accessible sources (e.g., ISO standards, government website URLs), and (3) metadata files containing source categories, processing timestamps, and quality control flags.
- **Code availability:** The source code for the models and experiments is available

at https://github.com/FAayn/EC_LLM.

- **Author contributions:** NaiPeng Wang led the manuscript writing and conducted the main experimental work. Linjie Tong and Longtao Zhang provided assistance with experiments and data processing. Chengxin Pang and Xiaoguang Zhu provided guidance on theoretical framework and research direction.

Appendix A. Detailed Architectural Features

We include these architectural details for completeness and reproducibility. EC-LLM does not claim novelty in core transformer components; instead, our contribution lies in how these established architectures are adapted and evaluated under compliance-critical, long-context energy-carbon settings. The following sections document the architectural choices that enable EC-LLM’s strong performance in domain-specific tasks.

EC-LLM integrates a suite of advanced architectural features to improve performance and efficiency in natural language processing tasks:

A.0.0.1. *Transformer Decoder Architecture.* Utilizes self-attention to model long-range dependencies in sequential data, enabling sophisticated language understanding and generation.

A.0.0.2. *Grouped Query Attention (GQA).* Reduces computational complexity by grouping query vectors while sharing key and value projections, allowing for faster inference without sacrificing accuracy.

A.0.0.3. *SwiGLU Activation Function.* In the feedforward neural network (FFN) layer, EC-LLM employs the Swish-Gated Linear Unit (SwiGLU) activation. Compared to traditional activation functions like ReLU, SwiGLU introduces a gating mechanism that enhances the model’s capacity to represent complex patterns in data, thereby improving overall performance Shazeer (2020). The detailed mathematical formulation is provided in Appendix B. To compare the activation effects of SwiGLU and ReLU, Yang et al. conducted comparative experiments based on the Qwen2.5-7B architecture, with results consolidated from ablation experiments. SwiGLU reduces confusion by 14.6% compared to ReLU, primarily due to the dynamic filtering of information through the gating mechanism, which enhances nonlinear expression capabilities. SwiGLU increases the number of parameters by approximately 15% due to the bilinear transformation, but this is compensated for by group query attention (GQA). In a 1M-token code understanding task, SwiGLU achieves a 9.2% higher semantic recall rate than ReLU, as the smoothing of gradients mitigates the decay of long-range dependencies. As such, SwiGLU consistently outperforms ReLU and GELU in language modeling tasks, with particularly significant gains in low-resource tasks and long-context scenarios. Its computational overhead can be optimized through the Mixture-of-Experts (MoE) architecture or GQA, making the actual deployment cost manageable Yang et al. (2025b).

A.0.0.4. *Attention QKV Biases.* Learnable biases are added to the Query, Key, and Value projections in the attention mechanism. These biases help stabilize training and improve model adaptability across diverse input types. Although specific imple-

mentation details remain undisclosed, QKV biases are considered beneficial for enhancing robustness.

A.0.0.5. Rotary Positional Encoding (RoPE). To encode both relative and absolute positional information, EC-LLM employs RoPE, which applies rotational transformations to query and key vectors Su et al. (2024). By gradually increasing the RoPE base frequency to 10,000,000, the Qwen team solved the extrapolation bottleneck of long-sequence position encoding. Combined with the DCA mechanism, the model achieved an accuracy rate of 95.1% in the 1M token Passkey Retrieval task, with a 3.2-6.7-fold improvement in memory efficiency Yang et al. (2025b). RoPE’s base frequency is adjusted based on model size, allowing for flexible generalization to inputs of varying lengths— particularly important for long-context scenarios.

A.0.0.6. RMSNorm for Normalization. EC-LLM adopts Root Mean Square Normalization (RMSNorm) in place of traditional LayerNorm. RMSNorm offers higher computational efficiency and enhanced training stability, especially in large-scale training settings Zhang and Sennrich (2019). The detailed formulation is provided in Appendix B.

A.0.0.7. Activation Function Ablation. Under identical training settings on Qwen2.5-7B, SwiGLU achieved the best overall trade-off: BLEU-4 of 31.3 and factual accuracy of 86.2%, while maintaining competitive inference speed (1480 tok/s) and acceptable GPU memory usage (10.8 GB). In comparison, GELU reached BLEU-4 of 27.6 and factual accuracy of 82.1% (1420 tok/s; 10.1 GB), and ReLU reached BLEU-4 of 24.1 and factual accuracy of 78.3% (1520 tok/s; 9.2 GB). These observations support the choice of SwiGLU in EC-LLM.

Appendix B. Mathematical Formulations

B.1. SFT Objective

The learning objective is to minimize the negative log-likelihood of targets conditioned on inputs (teacher-forced next-token prediction):

$$\mathcal{L}_{\text{SFT}}(\theta) = \sum_{i=1}^N \sum_{t=1}^{T_i} -\log p_{\theta}(y_i^{(t)} \mid y_i^{(<t)}, x_i). \quad (\text{B1})$$

B.2. LoRA Updates

For a weight matrix W in the frozen base model, we parameterize a trainable low-rank update:

$$W \leftarrow W_0 + \Delta W, \quad \Delta W = AB, \quad A \in \mathbb{R}^{d \times r}, \quad B \in \mathbb{R}^{r \times d}, \quad r \ll d. \quad (\text{B2})$$

B.3. *Distillation Loss*

We minimize the KL divergence between the teacher and student distributions:

$$\mathcal{L}_{\text{Distill}}(\theta) = \sum_i \text{KL}(p_T(\cdot | x_i) \| p_\theta(\cdot | x_i)), \quad \mathcal{L}(\theta) = \mathcal{L}_{\text{SFT}}(\theta) + \lambda \mathcal{L}_{\text{Distill}}(\theta), \quad (\text{B3})$$

with λ a small positive coefficient.

B.4. *SwiGLU Activation*

The Swish-Gated Linear Unit (SwiGLU) activation is defined as:

$$\text{SwiGLU}(x, y) = x \cdot \sigma(y), \quad (\text{B4})$$

where σ denotes the sigmoid function. Here, x and y denote the pre-activation linear outputs of two parallel feed-forward transformations.

B.5. *RMSNorm*

RMSNorm normalizes input vectors using their root mean square value:

$$\text{RMSNorm}(x) = \frac{x}{\sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2 + \epsilon}} \quad (\text{B5})$$

Appendix C. Evaluation Metrics Definitions

We employ a rigorous evaluation protocol involving both lexical and model-judged metrics to quantify performance. Let the test set be $\mathcal{T} = \{(q_j, c_j, a_j)\}_{j=1}^M$. Given inputs $x_j = [\text{BOS}; c_j; q_j]$, the model predicts $\hat{a}_j = f_\theta(x_j)$.

C.0.0.1. *Lexical metrics.* We compute BLEU-4 and ROUGE to measure textual overlap. BLEU is defined as:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^4 w_n \log p_n\right), \quad \text{BP} = \begin{cases} 1 & \text{if } L_{\hat{a}} > L_a, \\ \exp\left(1 - \frac{L_a}{L_{\hat{a}}}\right) & \text{otherwise,} \end{cases} \quad (\text{C1})$$

where p_n is the modified n-gram precision, w_n are uniform weights, and $L_a, L_{\hat{a}}$ denote reference and hypothesis lengths. For ROUGE-L, we use the longest common subsequence (LCS)-based F-score:

$$\text{ROUGE-L} = \max_{\text{LCS}} \text{F-score}(\text{precision}, \text{recall}). \quad (\text{C2})$$

C.0.0.2. *LLM-judged metrics.* To capture semantic correctness, we define three scalar scoring functions using an external judge model: semantic completeness $S_{\text{sem}}(x_j, \hat{a}_j)$, factual accuracy $S_{\text{fact}}(x_j, \hat{a}_j)$, and answer similarity $S_{\text{sim}}(x_j, \hat{a}_j)$, each

normalized to $[0, 100]$. With thresholds $\tau_{\text{sem}}, \tau_{\text{fact}}, \tau_{\text{sim}}$, the pass-rates are calculated as:

$$\text{Pass}_{\text{sem}} = \frac{1}{M} \sum_j \mathbf{1}\{S_{\text{sem}}(x_j, \hat{a}_j) \geq \tau_{\text{sem}}\}, \quad (\text{C3})$$

$$\text{Pass}_{\text{fact}} = \frac{1}{M} \sum_j \mathbf{1}\{S_{\text{fact}}(x_j, \hat{a}_j) \geq \tau_{\text{fact}}\}, \quad (\text{C4})$$

$$\text{Pass}_{\text{sim}} = \frac{1}{M} \sum_j \mathbf{1}\{S_{\text{sim}}(x_j, \hat{a}_j) \geq \tau_{\text{sim}}\}. \quad (\text{C5})$$

We also report per-metric averages $\bar{S}_{\bullet} = \frac{1}{M} \sum_j S_{\bullet}(x_j, \hat{a}_j)$. This protocol quantifies both lexical fidelity and domain soundness.

Appendix D. Data Generation Prompts

To ensure reproducibility and transparency, we provide the system prompts used for dataset generation. The content-based question generation employs Chain-of-Thought (CoT) prompting to guide the teacher model through structured reasoning.

D.1. Content-Based Question Generation Prompt

The following system prompt is used for content-based question generation from policy documents and technical standards:

You are an expert in energy-carbon domain knowledge extraction.
Given a document segment, generate 3-5 domain-specific questions that require deep understanding of regulatory frameworks, technical standards, or policy implications.

Process:

1. Identify key concepts: regulatory requirements, technical specifications, compliance procedures, or policy implications.
2. Formulate questions that test:
 - Understanding of specific standards (e.g., GB/T, ISO, GHG Protocol)
 - Application of regulations to real-world scenarios
 - Calculation procedures for emissions or energy metrics
 - Compliance verification steps
3. Ensure questions are:
 - Grounded in the provided context
 - Require multi-step reasoning
 - Test domain-specific knowledge, not general knowledge

Output format: JSON array of questions with associated complexity level (basic/intermediate/advanced).

D.2. Template-Based Question Generation Prompt

For vocabulary-based generation, we use the following template system:

You are generating domain-specific questions using expert-curated templates.
Given a term from the IPCC Vocabulary Database or energy-carbon terminology

list, generate a scenario-based question that requires:

1. Understanding of the term's definition
2. Application in a specific industry context
3. Compliance with relevant standards or regulations
4. Multi-step reasoning to arrive at the answer

Template structure:

- Context: [Policy/Standard/Regulation X]
- Entity: [Industry/Company Type Y]
- Condition: [Specific Scenario Z]
- Question: How should [Y] apply [term] under [X] when [Z]?

Ensure the generated question is specific, actionable, and requires domain expertise to answer correctly.

Appendix E. Human Evaluation Guidelines

To ensure consistency in human evaluation, we provided annotators with detailed scoring rubrics. Three domain experts assessed model outputs based on three dimensions using a 5-point Likert scale. The evaluation protocol achieved a Spearman correlation of 0.82 between human ratings and LLM-judged scores for factual accuracy, confirming the validity of our automated assessment protocol.

Appendix F. List of Covered Standards and Policies

Our dataset explicitly covers a wide range of national and international standards critical for energy-carbon management. The corpus includes international standards such as ISO 14064-1/2/3 for greenhouse gas accounting, ISO 14067 for carbon footprint of products, the GHG Protocol Corporate Accounting and Reporting Standard, and IPCC Guidelines for National Greenhouse Gas Inventories including both the 2006 version and 2019 Refinement.

National standards from China include GB/T 32150-2015 providing general guidelines for GHG emissions accounting and reporting of industrial enterprises, GB/T 24040/24044 covering environmental management and life cycle assessment, and the National Carbon Emissions Trading Management Rules from 2021. The dataset also incorporates regional policy documents, corporate case studies from leading energy sector companies, and analytical reports from governmental and non-governmental organizations focused on carbon emissions and energy transition.

Appendix G. Failure Case Analysis

While EC-LLM outperforms baselines across most scenarios, it still exhibits limitations in specific edge cases. The following example illustrates a failure mode where the model defaulted to higher-level policy information due to insufficient retrieval granularity.

Question: "What is the specific subsidy rate for distributed photovoltaics in Jinzhai County, Anhui Province in 2024?"

EC-LLM Answer: "According to the Anhui Provincial policy, the subsidy is 0.3 RMB/kWh. Specific county-level adjustments may apply but are not detailed in the provincial document."

Table E1. Scoring rubric provided to domain experts for human evaluation.

Metric	Score	Criteria Description
Factual Accuracy	5 (Perfect)	The answer is entirely factually correct, citing specific standards or policies accurately without any hallucination.
	3 (Acceptable)	The core information is correct, but minor details such as specific clause numbers are vague or missing. No misleading errors.
	1 (Fail)	Contains critical factual errors, applies wrong emission factors, or misinterprets regulations such as confusing Scope 1 and Scope 2 emissions.
Semantic Completeness	5 (Comprehensive)	Covers all aspects of the user’s query, including background, methodology, and necessary caveats.
	3 (Partial)	Answers the main question but misses relevant context or edge cases.
	1 (Incomplete)	Misses the core intent of the question or provides a generic, non-domain-specific response.
Answer Similarity	5 (Highly Similar)	The answer closely matches the reference in structure, terminology, and key points.
	3 (Moderately Similar)	The answer covers similar content but differs in organization or emphasis.
	1 (Dissimilar)	The answer diverges significantly from the reference in content or approach.

Issue: The model defaulted to the provincial policy because the specific county-level document for Jinzhai County was updated only one month prior to training and was not retrieved effectively due to low semantic overlap with the query. This highlights the dependency on retrieval granularity and the update frequency of the knowledge base, motivating our future work on continual learning pipelines to keep the model synchronized with rapidly evolving regional policies.

Another failure mode occurs when questions require integration of information from multiple disparate sources that are not co-located in the retrieval corpus. In such cases, EC-LLM may provide partial answers that cover only one aspect of a multi-faceted query, demonstrating the need for enhanced multi-hop reasoning capabilities in future iterations.

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